

Young Children's Analogical Reasoning: The Role of Immediate Feedback

Journal of Psychoeducational Assessment
2014, Vol. 32(5) 417–428
© 2013 SAGE Publications
Reprints and permissions:
sagepub.com/journalsPermissions.nav
DOI: 10.1177/0734282913515442
jpa.sagepub.com



Shuhui Chiu¹ and Patricia A. Alexander^{2,3}

Abstract

In this study, a computerized measure, Interactive Analogical Measure (IAM), was developed and used to assess young children's ability to reason analogically. The IAM was equipped to provide corrective feedback and the effects of that feedback were tested for experimental and control groups. A group of 5-year-olds ($N = 80$) participated in the study. Children were randomly assigned to the experimental or control group and the IAM was similarly implemented with the exception of the feedback indicators. There were significant differences between the experimental and control groups for overall performance. Data suggested that the effect of immediate, corrective feedback emerged very early during the test and manifested a cumulative effect across the entire test. It seems that an interactive measure provides meaningful problem-solving experience for young children.

Keywords

analogical reasoning, children, feedback, interactive measure

Analogical reasoning enables children to make inferences about relationally similar phenomena across contexts, which enhances young children's capability for transfer of learning and schema abstraction (Gentner & Markman, 1997; Goswami, 2001). Through the process of reasoning analogically, young children explore the salient features of a less familiar object, idea, or action (target) and seek to make sense of it by drawing similarities to an object, idea, or action that is more familiar (source). In so doing, the children's perceptive abilities may be enhanced and their conceptual understanding of both the target and the source may be augmented. In effect, such relational reasoning allows children to engage in assimilation and accommodation—abilities rudimentary to conceptual development (Piaget, 1952).

Over many decades, the significance of analogical reasoning has been repeatedly documented (e.g., English, 2004; Goswami, 1991). Specifically, analogical reasoning has been shown to be foundational to language development (Goswami & Mead, 1992), and conceptual change (e.g., Blanchette & Dunbar, 2002), key to scientific learning (e.g., Glynn & Takahashi, 1998), core to knowledge transfer (e.g., Alexander & Murphy, 1999), and a marker of creative and critical thinking (e.g., Alexander & the Disciplined Reading and Learning Research Laboratory, 2012;

¹National Taichung University of Education, Taiwan, R.O.C.

²University of Maryland, College Park, USA

³University of Auckland, New Zealand

Corresponding Author:

Shuhui Chiu, National Taichung University of Education, 402 No. 140 Minshen Rd. Taichung, Taiwan, R.O.C.
Email: shuhui@ms3.ntcu.edu.tw

Mumford, Medeiros, & Partlow, 2012). In recent years, empirical studies have also determined that analogical reasoning is manifest in young children's free play (Chiu & Tron, 2004) and is a predictor of preschoolers' learning effectiveness (Abdellatif, Cummings, & Maddux, 2008; Tunteler & Resing, 2007).

However, there is still much to learn about the nature of analogical reasoning in young children. The assessment issue, in particular, warrants more attention within the empirical research. If researchers are to understand the nature of analogical reasoning in young children, they must by necessity find tasks or measures that validly and reliably bring this way of thinking to the surface. When one is dealing with very young children who may be less verbally facile or whose experiences and associated conceptual knowledge are predictably limited, concerns over assessment are paramount. Often, when young children do not understand what is expected from them in a task, they may not solve the task in ways adults expected (Hessels, Vanderlinden, & Rojas, 2011; Shepard, Kagan, & Wurtz, 1998). Even if young children may understand the goal of a task, they may not be able to retain focus on that goal due to wavering motivation (Garris, Ahlers, & Driskell, 2002).

For this reason, we sought to embed the assessment of analogical reasoning for young children within a game-like environment that we would circumvent the issues of unclear goals and unmotivating conditions. We equipped this computerized game-like environment by providing these young children with immediate correct/non-correct feedback. Our intention was to empirically examine whether the inclusion of such immediate feedback enhanced these young children's analogical reasoning performance by contributing to goal refinement or maintaining motivation. To better understand the configuration of this game-like environment, we first overview more traditional assessment of analogical reasoning in young children

Traditional Assessments

In general, studies of analogical reasoning in young children have been broadly concerned with the application of such thinking processes in the solution of problems presented in narrative form or structured as classical analogy problems. In narrative form, children are presented a problem-solution posed as a story (e.g., Goldilocks and the Three Bears, Goswami, 1995) and are expected to use that information in solution of an unfamiliar problem (e.g., determining the relative magnitude of temperature).

In classical analogies of the form A:B::C:D, children may be given the source pair (i.e., A:B) and some portion of the corresponding target pair (e.g., C:?). Children then select the best term from among options that creates relational correspondence between the first and second pairings. In an attempt to reduce the abstractness and enhance the motivation of such classical analogy problems, researchers have sought to use very familiar objects (e.g., piggybanks or trucks, Crisafi & Brown, 1986) or to cast the problem-solving task in a game-playing context (White & Alexander, 1986).

Whether an analogy problem is posed as a story or in the form of A:B::C:?, assessments of analogical reasoning in young children have faced several criticisms. First, there is the repeated concern that the linguistic demands of the narrative forms may well underestimate analogical reasoning capacities in children who are still developing verbal facilities (Goswami & Brown, 1990). In regard to classic analogies, children may be unfamiliar to the A:B::C:D structures. For this reason, it is difficult enough for the children to identify the within-group relation (i.e., A:B or C:? lower-level relation), let alone the more complex higher-level relation (i.e., A:C; Singer-Freeman, 2005).

Furthermore, taking tests has not been a frequent occurrence in young children's lives. Young children have not been socialized in such aspects. Therefore, children may not recognize the goals of the assessment or understand what constitutes appropriate actions within this unfamiliar

situation (Hessels et al., 2011; Shepard et al., 1998). In addition, during traditional analogical testing, children are not provided with any corrective feedback; neither are they led to understand the purpose of taking the test from answering questions (Singer-Freeman & Bauer, 2008). Under such circumstances, children who may possess the cognitive abilities and conceptual understanding to respond correctly may fail to do so because they do not know the rules of the game.

Finally, there are motivational factors to consider. For instance, a traditional test may be uninteresting to children, who consequently do not fully engage in the assessment task. This is why a number of researchers of analogical reasoning in young children have turned to game-playing situations or enticing narrative problems to spark and maintain interest (Crisafi & Brown, 1986; Goswami, 1995; White & Alexander, 1986). Yet, finding the right content and structure to capture and maintain young children's interest throughout assessment remains a serious concern.

The Interactive Analogical Measure (IAM): A Computerized Game-Like Alternative

To address the aforementioned concerns and benefits of prior analogy assessments, the IAM was devised. Building on the positive features of prior assessments, such as familiar and interesting content, the IAM preserved the advantages of more traditional analogical testing with young children (e.g., well-investigated processing structure). Two other features of the IAM are especially noteworthy and pertinent to the current investigation. First, the IAM presented a task to children via the computer. Because of the computer-delivered nature of the assessment, we could provide immediate corrective feedback item-by-item. Second, this computerization allowed for the inclusion of tracking tools (e.g., eye tracking) that are not possible for paper-and-pencil measures.

Within the field of analogical research (Cheshire, Ball, & Lewis, 2005; Siegler & Svetina, 2002) and the general educational literature (Hattie, 2008), there is evidence that feedback is one of the most powerful factors to enhance learner performances. It has been argued that feedback provided by the experimenter enhances performance by maintaining the task goal in reasoners' working memory (Marcovitch, Boseovski, & Knapp, 2007) and by facilitating the creation of task-based mental models relevant to successful performance (Cheshire et al., 2005). With the guidance of task-based mental models, young children may apply more top-down control to selectively concentrate on relevant information, while ignoring trivial information (Kirkham & Diamond, 2003), and inhibit dominant impulse behavior to make a reasoned response (Diamond & Taylor, 1996).

The Current Study

Overall, based on the extant literature dealing with psychoeducational assessment, especially among young populations, we can hypothesize that the presence of immediate corrective feedback incorporated within a cognitive task, such as the IAM, could afford certain opportunities for the child and for the researcher. For the child, the inclusion of immediate corrective feedback could serve as relevant information as to the nature of the task, and, potentially, the attributes of a "correct" response to that task. For the researcher, the systematic inclusion of immediate feedback within an experimental study could provide an opportunity to investigate the effects of such task information on performance—not only overall but also as it unfolds across the task.

In effect, even if there is sufficient analogy-specific and general literature to hypothesize that feedback should contribute to significantly better analogy performance among experimental (with feedback) versus control (without feedback) participants, the literature is less clear on the response patterns that may contribute to that omnibus effect. For one, it is not clear whether the influence from feedback would be immediate, manifesting in the first few items of the IAM, or

would require time to develop, manifesting later in test performance. If, on the one hand, feedback contributes to more immediate understanding of goals of the problem-solving task (Hessels et al., 2011; Shepard et al., 1998), then we would expect to see changes in performance within the first few items of the interactive measures. On the other hand, if the benefits accrued from feedback take time to develop (if they prove beneficial at all), then it is conceivable that changes in performance would not be evidenced until later in the analogy task. In fact, this latter performance pattern is what was reported in several studies of analogical reasoning among children where feedback was provided orally by the experimenters (Cheshire et al., 2005; Siegler & Svetina, 2002).

Thus, given the importance of analogical reasoning to conceptual development, the lingering issues in the assessment of such processes in young children, and the potential effects of providing immediate and informative feedback to analogical performance, we designed the current investigation. Specifically, we designed an experimental study to address the following research questions:

- Is there a difference in performance accuracy on the IAM between those children given immediate, corrective feedback and those receiving no feedback?
- Is the effect for feedback immediately and consistently evident in children's performance of the IAM or is that effect more apparent in later items of the analogy task?

This study may help researchers and practitioners understand (a) how young children approach classic psychoeducational, analogical reasoning tasks, as well as (b) how feedback is reflected in children's performance, and correspondingly, assessments of their analogical reasoning abilities. Our findings may be useful for researchers who seek to design psychometrically sound psychoeducational assessment tools for young children.

Method

Participants

The participants for this research came from four daycare centers located in an urban city of central Taiwan. Parental permission was obtained for 80 of the 5-year-olds attending the centers (ages between 5.0 to 5.95 years; mean age = 5.51; $SD = 0.30$). All children were Taiwanese nationals and 42 were males. The 80 children from 14 classes were randomly assigned to either the experimental group (with feedback) or control group (without feedback) with consideration of their classroom placements and gender. For example, if possible, half of the males and females from the same class were randomly assigned to the experimental group for the reason that curriculum could vary among classes. After block randomization, the experimental group had a mean age of 5.46 ($SD = 0.30$), while the average age of the control group was 5.57 ($SD = 0.30$). The groups had comparable numbers of boys (21) and girls (19), and the majority of the children (84%) had experience operating a computer or watching others operate a computer.

The IAM

Item construction. The content and formatting of the IAM drew on procedures and principles established in prior studies of analogical reasoning in young children (e.g., Crisafi & Brown, 1986; White & Alexander, 1986). Items of the IAM were established through expert validation by researchers in assessment and child development and by practitioners in early childhood education. Based on prior research with young children (e.g., Lindvall & Ibarra, 1979), it was determined that illustrations would function better in item construction than photographic representations

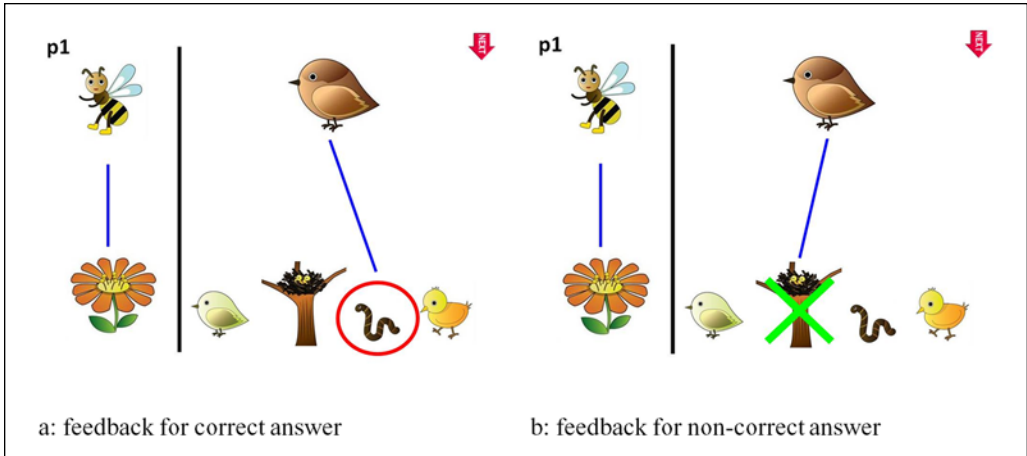


Figure 1. A sample item from interactive analogical test displaying the indicators for a correct and non-correct feedback.

in that they were not only simpler in form than photos but were also more stylized depictions of the object of interest. Once items using illustrations were constructed, they were evaluated by 5 early childhood practitioners who judged whether the objects depicted would be easily identifiable and familiar to 5-year-olds. Based on the feedback received, minor modifications in the drawings were made to produce the final set of IAM items.

Item content and formatting. The IAM was a 30-item multiple-choice test with an additional 3 non-scored practice items, which were displayed in a fixed order. These practice items were used to familiarize the children with the tool. The IAM covered analogical relations pertaining to associations that appear in daily lives, such as food, dwellings, clothing, occupations, transportation, leisure, nature, size, quantity, length, and whole-part. Each conceptual idea was represented by 2 items; for example, items related to dwelling were bird:nest::dog:? and spider:web::bee:?

As shown in Figure 1, in displaying the items for the IAM, we drew on the format for a *link-it* game with which the children were familiar. Specifically, for the problem bird: nest::dog:?, the source pairing (A:B; bird:nest) was displayed vertically on the computer screen and linked by means of a line to suggest a conceptual association. On the right hand of the screen, an image (C; dog) was placed toward the top, parallel with the “A” term in the problem. Four response options then were displayed vertically beneath the C term. Specifically, there was the correct answer (e.g., doghouse), a response within the same conceptual category (i.e., same; another dog), an object that was in an associated but conceptually distinct category (i.e., associated; cat as animal but not dog), and a more particular object that was within the target conceptual category (i.e., subcategory; bone as object associated with dog). The four categories of the answers were designed with reference to the principles proposed by Goswami and Brown (1990) for classical analogical testing in young children. We also placed a bold dark line to partition the examples in the previous pair from the question and selection items of the next pair. Overall, the design of the problem and its specific physical layout were intended (a) to visually establish for the children what the components of the given analogy problem were and (b) to spatially distinguish between the givens and the response options.

Feedback condition. This research adopted the experimental method to explore the effects of feedback on analogical reasoning performance in that the experimental group received feedback

during test performance, while the control group did not. The computer program of IAM was the same for the experimental and control groups with the exception that no immediate feedback was provided for children in the control group. Specifically when a child in the feedback condition clicked on the correct response option answer, a “o” appeared on the screen accompanied with a “clapping hand” sound effect (Figure 1a). Conversely, when the child clicked on any of the other three distractors (i.e., same category, associated category, or subcategory), an “X” appeared on the screen followed by a beeping sound (Figure 1b). These indicators “o” and “X” are well known in Chinese culture representing correct and incorrect answers. No such indicators appeared on the screen when children in the control group clicked on any of the response options—correct or otherwise.

Psychometric properties. We used the Kuder-Richardson reliability index to measure the consistency of the children’s answers on the IAM. For this analysis, we examined the responses for sixty 4.42- to 7.42-year-olds, comparable in background to the study sample. Results showed that the children’s answers for the 30 items had high degree of internal consistency ($KR_{20} = .90$; all with feedback). The accuracy rates of the 4-, 5-, 6-, and 7-year-olds were .32, .52, .67, and .81, respectively. We also looked at the test-retest validity of the IAM by administering and re-administering the test to 18 children after 2-week delay. It was our assumption that some analogical development may have occurred over that time, but the children’s performance remained relatively stable, as was the case ($r = .67, p < .001$).

Procedure

In the feedback and non-feedback condition, the tests were similarly implemented with the exception of the feedback indicators. In a set-off space within the daycare center, a researcher met individually with a child. The researcher began by telling the child, “You are invited to play a game. I will tell you how to play, and after that, you will play it by yourself.” Children were told to try “your best, and see whether you can find the correct picture.” We used the practice items to reinforce this procedure with the children before they were presented with the actual test items. For example, in a practice question, the researcher stated that

In the drawings on the left side, you can see [body] and [cloth] linked together (researcher points to each of the “A” and “B” pictures). This means they are a pair. Please pick one from the four pictures (indicating the four response options) to make a pair with the [head] (pointing to the “C” term), just like the pair at the left-hand side.

If the child selected the correct option, the researcher responded, “You are right.” The researcher then explained, “Our body wears clothes just like our head wears hat.” If the child gave an incorrect answer, the researcher responded, “Oops! The correct answer is hat.” The researcher then stated the reason. Following the same procedure, the researcher led the child through three practice questions to familiarize the child with the operation of the test instrument and the goal of the test questions.

After demonstrating the practice items, the actual test began. For each item, the researcher stated, “This is [A], this is [B], and this is [C] (researcher points to each of the “A,” “B,” and “C” pictures), which one is the correct answer? (a finger sweep across the four response options)” The researcher let the child use the mouse to click on the picture she or he wanted to select and refrained from giving any feedback on the accuracy of that choice.

In the experimental group, the computer provided instant feedback as to the correctness or incorrectness of each response in the manner indicated earlier. When a child gave an incorrect answer, the research asked whether she or he would like to try again. Generally, the child would

spontaneously try again until the correct answer was chosen (i.e., a situation like playing a computer game). If the children did not wish to try until a correct answer was achieved, they may click the “next” button and move to the next item. However, no such case appeared in the current study. Therefore, the feedback in each test item was not limited to correct/incorrect information received for the first attempt. Despite this re-selection component, only the initial response was considered in data analysis. In the control group, the children received no indication of item correctness. After each selection, the children were instructed to click on the “next” button and move to the next item. Thus, children in the control group made one choice per item.

Data Analyses

Kuder-Richardson reliability index was used to measure the consistency of the children’s answers on the IAM. Specifically, children’s performance was measured by calculating their average accuracy rate. An average accuracy rate of .50 indicated that a given child had responded accurately to 15 items of the 30-item measure. To examine the effect of giving immediate feedback, we compared average accuracy rates of the experimental and control groups under the assumption that any differences in these rates would be associated with the feedback received by the treatment group.

Furthermore, to determine the point at which the effect of giving immediate feedback manifests in test performance, we examined performance of the experimental and the control groups for test segments. At the item level, an accuracy rate of .50 indicated that half of the children had responded accurately on that item. However, because performance at item level was expected to fluctuate across the test as a consequence of repeated exposures, we chose to analyze response accuracy for six test segments, consisting of 5 items each. In this way, we could detect any trend in response accuracy in a more reliable manner. We determined the accuracy rate for these test segments by taking the average performance for the 5 items in that segment. For example, an accuracy rate of .80 for a given segment meant that the child identified the correct option for 4 of the 5 items.

Results and Discussion

Performance of the Experimental and Control Groups

Performance consistency. In the current study, the internal consistency for the 5-year-old participants was .80 for the experimental group and .67 for the control group. Based on these data, it appeared that the lack of immediate feedback contributed to less response consistency for the control group across the 30 test items. This finding also supports our hypothesis that under non-feedback condition, children may fail to respond as consistently across the test because they fail to grasp the reasoning required to identify the correct answer and thus may respond more inconsistently.

Overall performance. In general, the performance data from the IAM indicated that the 5-year-old children who participated in this study demonstrated the ability to think analogically to a reasonable level on the IAM, whether they received immediate, corrective feedback or not. Specifically, the 80 children displayed an average accuracy rate of .58 ($SD = 0.16$). That is, they responded accurately to more than half of the IAM items—a level of performance for all participants well above chance (i.e., between 7 and 8 items for the 30-item measure, $t = 9.22$, $p < .001$). The children’s performance accuracy for the 30 items ranged from a low of .27 (near chance level) to a high of .93. It seems that there were still some children whose abilities to infer the relational similarities between the source and target pairs of the items were not well developed.

Table 1. Performance Accuracy of the Experimental and Control Groups Across the Entire Test.

Item	Experimental		Control		Difference	Cohen's <i>d</i>	<i>t</i>	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
1-5	.51	.29	.46	.22	.05	.20	0.88	.382
6-10	.73	.17	.67	.24	.06	.29	1.28	.206
11-15	.52	.23	.41	.19	.11	.50	2.22	.030*
16-20	.68	.24	.60	.24	.08	.33	1.48	.144
21-25	.62	.24	.51	.24	.11	.46	2.07	.043*
26-30	.67	.27	.56	.22	.12	.47	2.11	.038*
Entire test	.62	.17	.53	.14	.09	.55	2.48	.015*

Note. *n* = 40 for each group.

**p* < .05.

Between group performance. As shown in Table 1, the children in the experimental group had a significantly higher accuracy rate ($M = 0.62$, $SD = 0.17$) for the 30 analogy items than the children in the control group ($M = 0.53$, $SD = 0.14$, $t = 2.48$, $p = .015$). The effect size measured in Cohen's *d* was medium to large ($d = .55$). That is children given immediate feedback performed much better than those receiving no feedback.

The feedback effect reported here may arise both from “hot” and “cold” cognitive mechanisms. In the “hot” cognitive aspect, children may be more enthusiastic about solving the analogical problems when feedback is given immediately after they respond (Garris et al., 2002). In the “cold” cognitive aspect, the information conveyed in the feedback may contribute to the understanding of task goal (Hessels et al., 2011; Shepard et al., 1998) and/or facilitate the creating of task-based mental models (Cheshire et al., 2005) that guide subsequent performance. Of course, it is also possible that the performance gains of the experimental group may also come from the children's successive attempts after they responded incorrectly. However, we found that when the children initially responded incorrectly, their second attempts were often correct (92%), thus, minimizing the practice.

Feedback Effect at Item Level

Item performance. As seen in Figure 2, for the 1st 6 items, the performance of the two groups was not very dissimilar. Specifically, the children in the experimental group preformed slightly better for Items 1 and 5 but slightly worse for Items 3 and 6, as compared with the children in the control group. This indicated that the children in the experimental and the control group were reasoning comparably at the beginning of the test.

However, after the sixth item, the experimental group performed constantly better. In Figure 2, the gray area marks the degree of differential performance between the experimental and control groups. As indicated by the dashed line, the differences between experimental and control groups were noticeably greater after the sixth item—a pattern that persisted across most of the remaining items. This pattern suggests that the effect of giving immediate, corrective feedback emerged very early during the test. We suspect that the rapid emergence of the feedback effect comes from the understanding of task goal (Hessels et al., 2011; Shepard et al., 1998); that is, children quickly came to understand what the task required of them and how they were to respond.

Segment performance. As seen in Figure 3, the performance trajectories of the experimental group and in the control group followed rather parallel paths, with the experimental group consistently

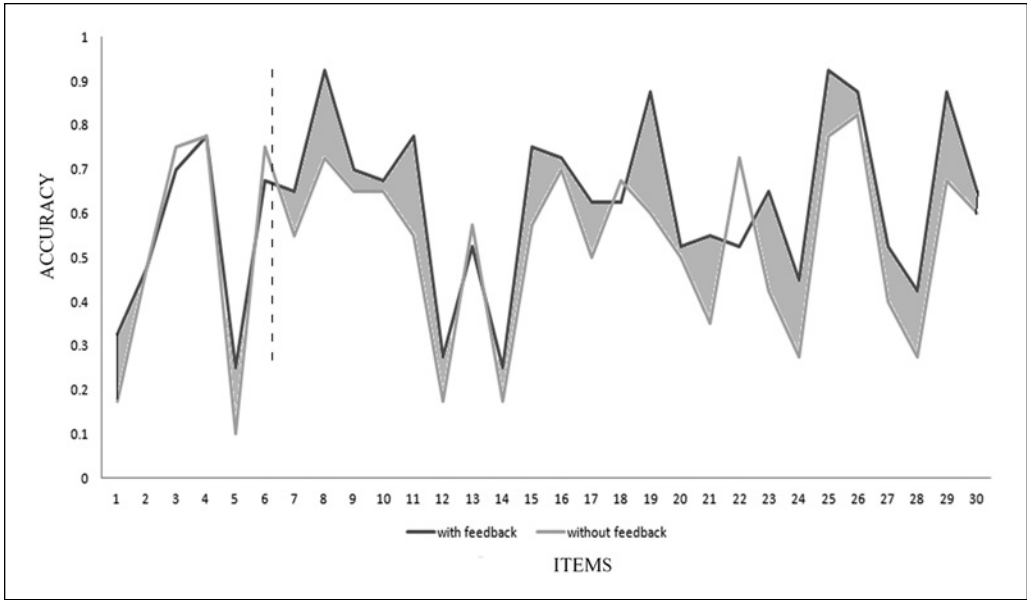


Figure 2. The differential performance accuracy for experimental and control groups (Gray Area) across the 30 items of the IAM.

Note. IAM = Interactive Analogical Measure.

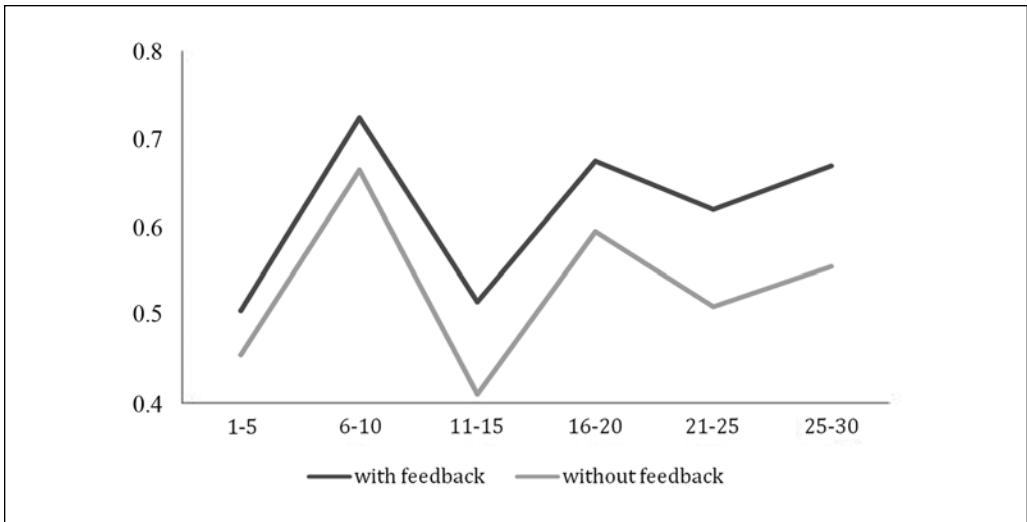


Figure 3. Performance trajectories for the experimental and control groups for six segments of the IAM.

Note. IAM = Interactive Analogical Measure.

outperforming the control group. However, those paths diverged to a somewhat greater degree in the later test segments. As seen in Table 1, the disparities of the two groups enlarged from 5% to 12% from the first segment to the last test segment. The accuracy rates of the two groups did not differ significantly for the first and second segments ($ts < 1.28$, $ps > .206$, $ds < .29$), but differed significantly for the fifth and the last segments ($ts > 2.07$, $ps < .05$, $ds > .46$). These data suggest that the effect of immediate, corrective feedback may manifest a cumulative effect across the

entire test of IAM. The cumulative effect may indicate some degree of learning consistent to what was reported in previous studies (Cheshire et al., 2005; Siegler & Svetina, 2002). That is, by repeatedly receiving corrective information, children learn to inhibit their tendency to respond quickly to a given problem based on more superficial features and to attend to the deeper structure of the items (Cho, Holyoak, & Cannon, 2007; Richland, Morrison, & Holyoak, 2006).

Conclusions

In this study, we set out to test the viability of a computer-delivered interactive analogy measure for young children and to experimentally ascertain whether the incorporation of immediate, corrective feedback had a significant effect on children's performance. By comparing children's performance accuracy on the IAM between those children given immediate, corrective feedback and those receiving no feedback, we were able to explore the role the game-like feedback would play in the accuracy of the children's responses.

What we determined from the data was that children's performances within the entire test were more consistent when feedback was received than was not. Corrective feedback did, in fact, have an immediate effect on the children's performance accuracy, as evidenced by a rather quick rise in overall performance from the outset of the testing. We also witnessed an incremental improvement in performance accuracy across the entire assessment, suggesting children learn to engage in the processes needed for analogical problem solving. Our findings suggested that classic psychoeducational measurements might well underestimate young children's capabilities. An interactive measure, such as the IAM, which provides a game-like environment and which incorporates immediate corrective feedback, could enhance the meaningfulness of the activity for young children, and, thus, allow them to manifest analogical abilities that might not be revealed under more traditional test-like conditions.

In this study, we examined the feedback effect only for a limited age range. Whether children younger or older with various analogical reasoning abilities derive different benefits from the feedback is still uncertain. Thus, future studies that examine whether there is significant interplay between ability and feedback condition are warranted. In addition, in the current study, the performance of the experimental group may have benefitted from successive attempts following an incorrect response. Consequently, future studies could explicitly test for the effects of performance when successive response efforts are allowed or inhibited. Finally, considering the advantage of the digitalized nature of IAM, subsequent investigations could include eye tracking as a way to correlate young children's patterns of saccades and fixations with their performance outcomes under feedback or no-feedback conditions. Although we acknowledge the exploratory nature of this investigation and concede that there is still much more to be learned about the underlying mechanism of analogical reasoning, we consider the current study an important foray into that area of inquiry.

Acknowledgments

We thank the Taiwan NSC (NSC 99-2511-S-142-011-MY2) for funding this project.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The authors thank the Taiwan NSC (NSC 99-2511-S-142-011-MY2) for funding this project.

References

- Abdellatif, H. R., Cummings, R., & Maddux, C. D. (2008). Factors affecting the development of analogical reasoning in young children: A review of literature. *Education, 129*, 239-249.
- Alexander, P. A., & the Disciplined Reading and Learning Research Laboratory. (2012). Reading into the future: Competence for the 21st century. *Educational Psychologist, 47*, 1-22. doi:10.1080/00461520.2012.722511
- Alexander, P. A., & Murphy, P. K. (1999). Nurturing the seeds of transfer: A domain-specific perspective. *International Journal of Educational Research, 31*, 561-576.
- Blanchette, I., & Dunbar, K. (2002). Representational change and analogy: How analogical inferences alter target representations. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 28*, 672-685.
- Cheshire, A., Ball, L. J., & Lewis, C. N. (2005). Self-explanation, feedback and the development of analogical reasoning skills: Microgenetic evidence for a metacognitive processing account. In B. G. Bara, L. W. Barsalou, & M. Bucciarelli (Eds.), *Proceedings of the twenty-seventh annual conference of the cognitive science society* (pp. 435-441). Mahwah, NJ: Lawrence Erlbaum.
- Chiu, S., & Tron, M. G. O. (2004). Classroom discourse and mathematics reasoning: A three-year longitudinal study. In L. D. English (Ed.), *Mathematical and analogical reasoning of young learners* (pp. 75-96). Mahwah, NJ: Lawrence Erlbaum.
- Cho, S., Holyoak, K. J., & Cannon, T. D. (2007). Analogical reasoning in working memory: Resources shared among relational integration, interference resolution, and maintenance. *Memory & Cognition, 35*, 1445-1455.
- Crisafi, M. A., & Brown, A. L. (1986). Analogical transfer in very young children: Combining two separately learned solutions to reach a goal. *Child Development, 57*, 953-968.
- Diamond, A., & Taylor, C. (1996). Development of an aspect of executive control: Development of the abilities to remember what I said and to "Do as I say, not as I do." *Developmental Psychology, 29*, 315-334.
- English, L. D. (2004). Mathematical and analogical reasoning of young children. In L. D. English (Ed.), *Mathematical and analogical reasoning of young learners* (pp. 1-22). Mahwah, NJ: Lawrence Erlbaum.
- Garris, R., Ahlers, R., & Driskell, J. E. (2002). Games, motivation, and learning: A research and practice model. *Simulation & Gaming, 33*, 441-467.
- Gentner, D., & Markman, A. B. (1997). Structure mapping in analogy and similarity. *American Psychologist, 52*, 45-56.
- Glynn, S. M., & Takahashi, T. (1998). Learning from analogy-enhanced science text. *Journal of Research in Science Teaching, 35*, 1129-1149.
- Goswami, U. (1991). Analogical reasoning: What develops? A review of research and theory. *Child development, 62*, 1-22.
- Goswami, U. (1995). Transitive relational mappings in three- and four-year-olds: The analogy of Goldilocks and the Three Bears. *Child Development, 66*, 877-892.
- Goswami, U. (2001). Analogical reasoning in children. In D. Gentner, K. J. Holyoak, & B. N. Kokinov (Eds.), *The analogical mind: Perspectives from cognitive science* (pp. 437-470). Cambridge, MA: MIT Press.
- Goswami, U., & Brown, A. L. (1990). Higher-order structure and relational reasoning: Contrasting analogical and thematic relations. *Cognition, 36*, 207-226.
- Goswami, U., & Mead, F. (1992). Onset and rime awareness and analogies in reading. *Reading Research Quarterly, 27*, 152-162.
- Hattie, J. (2008). *Visible learning: A synthesis of over 800 meta-analyses relating to achievement*. New York, NY: Routledge.
- Hessels, M. G. P., Vanderlinden, K., & Rojas, H. (2011). Training effects in dynamic assessment: A pilot study of eye movement as indicator of problem solving behavior before and after training. *Educational and Child Psychology, 28*, 101-113.
- Kirkham, N., & Diamond, A. (2003). Sorting between theories of preservation: Performance in conflict tasks requires memory, attention and inhibition. *Developmental Science, 6*, 474-476.
- Lindvall, C. M., & Ibarra, C. G. (1979). *The development of problem-solving capabilities in kindergarten and first-grade children*. Pittsburgh, PA: Learning Research and Development Center, University of Pittsburgh.

- Marcovitch, S., Boseovski, J. J., & Knapp, R. J. (2007). Use it or lose it: Examining preschoolers' difficulty in maintaining and executing a goal. *Developmental Science*, 10, 559-564.
- Mumford, M. D., Medeiros, K. E., & Partlow, P. J. (2012). Creative thinking: Processes, strategies, and knowledge. *Journal of Creative Behavior*, 46, 30-47.
- Piaget, J. (1952). *The origins of intelligence in children* (M. Cook, Trans.). New York, NY: W. W. Norton.
- Richland, L. E., Morrison, R. G., & Holyoak, K. J. (2006). Children's development of analogical reasoning: In sights from scene analogy problem. *Journal of Experimental Child Psychology*, 94, 249-273.
- Shepard, L., Kagan, S. L., & Wurtz, E. (1998). *Principles and recommendations for early childhood assessments*. Washington, DC: National Education Goals Panel.
- Siegler, R. S., & Svetina, M. (2002). A microgenetic/cross-sectional study of matrix completion: Comparing short-term and long-term change. *Child Development*, 73, 793-809.
- Singer-Freeman, K. E. (2005). Analogical reasoning in 2-year-olds: The development of access and relational inference. *Cognitive Development*, 20, 214-234.
- Singer-Freeman, K. E., & Bauer, P. J. (2008). The ABCs of analogical abilities: Evidence for formal analogical reasoning abilities in 24-month-olds. *British Journal of Developmental Psychology*, 26, 317-335.
- Tunteler, E., & Resing, W. C. M. (2007). Effects of prior assistance in using analogies on young children's unprompted analogical problem solving over time: A microgenetic study. *British Journal of Educational Psychology*, 77, 43-68.
- White, C. S., & Alexander, P. A. (1986). Effects of training on four-year-olds' ability to solve geometric analogy problems. *Cognition and Instruction*, 3, 261-268.